

Grade 6 Accelerated
Unit 12
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

A map is set to a scale factor of $32 \text{ mm} = 1 \text{ km}$. Use this scale to answer questions 1-3.

1. Two mountains are 451.1 mm apart on the map. Create a proportion that could be used to determine the actual distance between the mountains.

$$\frac{32 \text{ mm}}{1 \text{ km}} = \frac{451.1 \text{ mm}}{x \text{ km}}$$

2. Use proportions to solve for the actual distance between the mountains. Round the answer to the nearest tenth.

$$32 \cdot x = 1 \cdot 451.1$$

$$x = 14.1 \text{ km}$$

$$\frac{32x}{32} = \frac{451.1}{32}$$

$$x = 14.096875$$

3. Using the scale from the map in question 1, if the distance between two towns is 25 kilometers, how many millimeters apart will the towns be on the map?

$$\frac{32 \text{ mm}}{1 \text{ km}} = \frac{x \text{ mm}}{25 \text{ km}}$$

$$32 \cdot 25 = 1x$$

$$800 = x$$

The towns will be 800 mm apart on the map.

In a construction zone, there are black and gray beams that are being used for a building that are proportional in size. The black beams are 5 feet wide and 16 feet long, and the gray beams are 12.5 ft long. Use this information to answer questions 4-5.

4. Fill in the remaining values for the proportion below.

$$\frac{5}{x} = \frac{16}{12.5}$$

5. Using the above proportion, determine the width of the gray beams.

$$5 \cdot 12.5 = 16x$$

The gray beam are 3.9 feet wide.

$$\frac{62.5}{16} = \frac{16x}{16}$$

$$3.90625 = x$$

Use the following information to solve problems 6-10. Suppose 8,800 people live in apartment buildings in Charlotte County, Florida. A sample of 310 people were polled asking them, if given the opportunity, would they move into a new house.

- 144 said yes.
- 150 said no.
- 16 were undecided.

6. Create a proportion to determine the total number of people who were undecided.

$$\frac{16 \text{ undecided}}{310 \text{ polled people}} = \frac{x \text{ undecided}}{8,800 \text{ people in city}}$$

$$16 \cdot 8,800 = 310x$$

$$\frac{140,800}{310} = \frac{310x}{310} \approx 454 \text{ people were undecided}$$

$$454.193584 = x$$

$$\frac{16}{310} = \frac{x}{8,800}$$

7. Create a proportion to determine the total number of people who said yes.

$$\frac{144 \text{ yes}}{310 \text{ polled}} = \frac{x \text{ yes}}{8,800 \text{ people}}$$

$$= 144 \cdot 8,800 = 310x$$

$$\frac{1,267,200}{310} = \frac{310x}{310} \approx 4,088 \text{ people said yes}$$

$$4,087.7419 = x$$

$$\frac{144}{310} = \frac{x}{8,800}$$

8. Create a proportion to determine the total number of people who said no.

$$\frac{150 \text{ no}}{310 \text{ polled}} = \frac{x \text{ no}}{8,800 \text{ people}}$$

$$= 150 \cdot 8,800 = 310x$$

$$\frac{1,320,000}{310} = \frac{310x}{310} \approx 4,258 \text{ people said no}$$

$$4,258.06452x$$

$$\frac{150}{310} = \frac{x}{8,800}$$

9. To the nearest whole number, determine how many people were undecided.

$$16 \cdot 8,800 = 310x$$

$$\frac{140,800}{310} = \frac{310x}{310} \approx 454 \text{ people were undecided}$$

$$x = 454.193584$$

10. To the nearest whole number, determine the number of people who said yes.

$$144 \cdot 8,800 = 310x$$

$$\frac{1,267,200}{310} = \frac{310x}{310} \approx 4,088 \text{ people said yes}$$

$$4,087.7419 = x$$